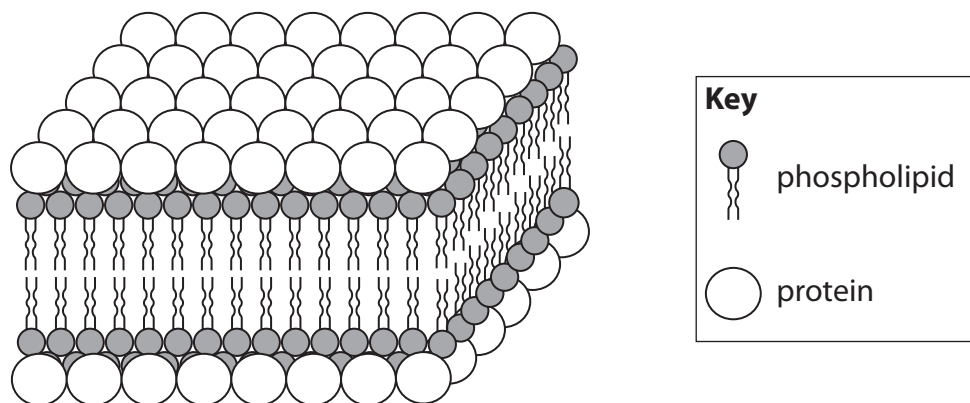


- 6 The fluid mosaic model of membrane structure was developed to explain the structure and properties of cell membranes.

(a) One model of membrane structure was developed by Davson and Danielli.

The diagram shows the Davson–Danielli model.



(i) Which describes a phospholipid?

(1)

- ☐ **A** hydrophilic fatty acid head and hydrophobic phosphate tails
- ☐ **B** hydrophilic phosphate head and hydrophobic fatty acid tails
- ☐ **C** hydrophobic fatty acid head and hydrophilic phosphate tails
- ☐ **D** hydrophobic phosphate head and hydrophilic fatty acid tails



- (ii) Compare and contrast the structure of the model shown in the diagram with that of the fluid mosaic model of membrane structure.

(4)

- (b) The cell membrane is important to the cell and if it gets damaged the cell could die.

- (i) Suggest how damage to the cell membrane could result in cell death.

(2)



- (ii) Cells can repair damage to the cell membrane.

This involves fusion of parts of the membrane around the edge of the damage.

Explain why parts of the membrane are able to fuse together to repair the damage.

(2)

- (iii) Once the membrane has fused together, other processes are needed to complete the repair of the membrane.

Suggest what else the cell needs to do to complete this repair.

(2)

(Total for Question 6 = 11 marks)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

